

# RAG Assignment

## **Q1. What problem does RAG solve that traditional LLMs cannot?**

RAG gives LLMs access to up-to-date external knowledge instead of relying only on training data.

## **Q2. Why do LLMs hallucinate, and how does RAG reduce hallucinations?**

LLMs may generate incorrect facts when uncertain. RAG retrieves relevant documents and grounds the response in them.

## **Q3. If an LLM already has billions of parameters, why does it still need external knowledge?**

Training data becomes outdated and cannot include every fact. External knowledge keeps answers current.

## **Q4. Compare a chatbot with and without RAG. Which one would you trust more and why?**

A RAG chatbot is generally more trustworthy because it answers using retrieved documents.

## **Q5. Can RAG completely eliminate hallucinations? Justify your answer.**

No. It reduces hallucinations but cannot remove them completely if retrieval or reasoning is poor.

## **Q6. Why is retrieving the right information as important as generating the right answer?**

Correct retrieval provides accurate context, allowing better generation.

## **Q7. Explain the complete workflow of a RAG system using a real-world example.**

User query → embedding → vector search → retrieve relevant chunks → add to prompt → LLM generates answer.

## **Q8. What role do embeddings play in a RAG pipeline?**

They convert text into vectors so semantic similarity search is possible.

## **Q9. Why can't we directly search documents using keywords in modern RAG systems?**

Keywords miss meaning; semantic search understands intent.

## **Q10. How does semantic search differ from keyword search?**

Semantic search matches meaning; keyword search matches exact words.

## **Q11. Why is chunking necessary before storing documents in a vector database?**

Chunking improves retrieval accuracy and fits model context windows.

## **Q12. What could happen if document chunks are too large or too small?**

Large chunks add noise; small chunks lose context.

## **Q13. Compare dense and sparse embeddings. Which for legal search?**

Dense embeddings capture meaning; sparse focus on exact terms. Legal search often benefits from a hybrid approach.

## **Q14. Why are vector databases preferred over relational databases for RAG?**

Vector DBs efficiently perform similarity search on embeddings.

## **Q15. Explain cosine similarity.**

It measures similarity between vectors; higher similarity means more relevant documents.

## **Q16. What factors influence retrieval quality?**

Embedding quality, chunking, retrieval method, document quality, and query formulation.

## **Q17. How does poor retrieval affect answers?**

The LLM receives weak context and may produce incorrect or incomplete answers.

**Q18. Importance of prompt engineering in RAG?**

Good prompts clearly instruct the model to use retrieved context and answer accurately.

**Q19. How is context injected?**

Retrieved document chunks are appended to the prompt before generation.

**Q20. Why prefer RAG over fine-tuning for changing info?**

Updating documents is easier and cheaper than retraining a model.

**Q21. Compare RAG and fine-tuning.**

RAG adds external knowledge; fine-tuning changes model behavior. Use RAG for changing knowledge and fine-tuning for specialized behavior.

**Q22. Can RAG work without a vector DB?**

Yes, using keyword search or databases, though vector databases usually give better semantic retrieval.

**Q23. Advantages/limitations of FAISS?**

FAISS is fast and scalable but lacks built-in persistence and distributed features.

**Q24. Build RAG for university docs.**

Index university policies, notes, and regulations, retrieve relevant content, then generate answers.

**Q25. Design RAG for hospital guidelines.**

Store medical guidelines securely, retrieve relevant protocols, and let the LLM answer with citations.

**Q26. How improve customer support?**

It provides fast, accurate, document-based responses to customer questions.

**Q27. Security/privacy challenges?**

Protect sensitive data with authentication, encryption, access control, and audit logs.

**Q28. Evaluation metrics?**

Retrieval precision/recall, answer accuracy, faithfulness, latency, and user satisfaction.

**Q29. Reasons for wrong answers despite retrieval?**

Retrieved context may be irrelevant, incomplete, ambiguous, or the prompt/model may interpret it poorly.

**Q30. Future of RAG?**

RAG will become more accurate, multimodal, agentic, personalized, and real-time.